

Answer key | Lesson 9 - Perimeter and area

Mild

1) Perimeter AND area of each rectangle

- a) Perimeter **14 cm**, Area **12 cm²**
 b) Perimeter **22 cm**, Area **30 cm²**
 c) Perimeter **20 cm**, Area **16 cm²**

2) Complete the table

L	W	P	A
5	3	16	15
8	4	24	32
7	5	24	35
9	6	30	54

Row 2: length 8. Row 3: width 5.

Spicy

1) Find the missing side length

- a) $28 \div 2 - 9 = 5$ cm
 b) $54 \div 6 = 9$ cm
 c) $46 \div 2 - 13 = 10$ cm
 d) $84 \div 7 = 12$ cm

2) Word problems

- a) Perimeter **346 m**, Area **7 140 m²**
 b) Each side is 9 m, so perimeter **36 m**

Hot

1) Same area, different perimeter

1×36 : perimeter 74 cm
 2×18 : perimeter 40 cm
 3×12 : perimeter 30 cm
 4×9 : perimeter 26 cm
 6×6 : perimeter 24 cm
 Smallest: **6 cm $\times$ 6 cm (24 cm)**

2) Design challenge

10 m $\times$ 10 m, area 100 m². Length plus width must be 20. The closer to a square, the bigger the area ($11 \times 9 = 99$, $12 \times 8 = 96$).

Reasoning

False. 1×9 and 5×5 both have perimeter 20 cm, but their areas are 9 cm² and 25 cm².